

Python Basics Homework Assignment

Assignment Title

Python Fundamentals: Variables, Data Types, Lists, Dictionaries, Operators, and Conditions

Instructions

- Answer all questions.
 - Write Python code wherever required.
 - Use meaningful variable names.
 - Add comments to explain your logic.
 - Test your programs with different inputs.
 - Total Marks: **100**
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Section A: Multiple Choice Questions (10 Marks)

Q1. Which of the following is a valid variable name in Python?

- a) 2name
- b) my-name
- c) my_name
- d) class

Marks: 1

Q2. What will be the data type of 3.14?

- a) int
- b) float
- c) string
- d) boolean

Marks: 1

Q3. Which operator is used for exponentiation?

- a) ^
- b) *
- c) //
- d) **

Marks: 1

Q4. Which data structure stores data in key-value pairs?

- a) List
- b) Tuple
- c) Dictionary
- d) Set

Marks: 1

Q5. What is the output?

None

```
x = 10
```

```
y = 20
```

```
print(x > y)
```

- a) True
- b) False
- c) 10
- d) Error

Marks: 1

Q6. Which function is used to find the length of a list?

- a) size()
- b) count()
- c) len()
- d) length()

Marks: 1

Q7. Which of the following is mutable?

- a) String
- b) Integer
- c) List
- d) Float

Marks: 1

Q8. What will be the output?

None

```
print(10 % 3)
```

- a) 0
- b) 1
- c) 3
- d) 10

Marks: 1

Q9. Which keyword is used for an alternative condition?

- a) otherwise
- b) elseif
- c) elif
- d) else if

Marks: 1

Q10. Which operator checks equality?

- a) =
- b) ==
- c) !=
- d) >=

Marks: 1

Section B: Short Answer Questions (20 Marks)

Q11. What is a variable? Give an example.

Marks: 2

Q12. Differentiate between Integer and Float with examples.

Marks: 2

Q13. What is the difference between a List and a Dictionary?

Marks: 3

Q14. Explain the following operators with examples:

- Arithmetic Operators
- Relational Operators
- Logical Operators

Marks: 5

Q15. What is the purpose of conditional statements in Python?

Marks: 3

Q16. Write the syntax of:

1. if statement
2. if-else statement
3. if-elif-else statement

Marks: 5

Section C: Programming Questions (40 Marks)

Q17. Create variables to store:

- Your name
- Age
- City

Print them in a formatted sentence.

Marks: 5

Q18. Write a program to:

- Take two numbers as input
- Perform addition, subtraction, multiplication, and division
- Display all results

Marks: 5

Q19. Create a list containing five favorite fruits.

Perform the following:

1. Print the list
2. Add one fruit
3. Remove one fruit
4. Print the updated list

Marks: 5

Q20. Create a dictionary to store student information:

None

```
{  
    "name": "John",  
    "age": 20,  
    "course": "Python"  
}
```

Perform the following:

1. Print all values
2. Add a new key called "city"
3. Update age
4. Print the updated dictionary

Marks: 5

Q21. Write a program to check whether a number is positive, negative, or zero.

Marks: 5

Q22. Write a program to find the larger of two numbers using an if-else statement.

Marks: 5

Q23. Write a program that checks whether a student has passed or failed.

Criteria:

- Marks $\geq 40 \rightarrow$ Pass
- Marks $< 40 \rightarrow$ Fail

Marks: 5

Q24. Write a program to determine whether a person is eligible to vote.

Criteria:

- Age $\geq 18 \rightarrow$ Eligible
- Otherwise $\rightarrow$ Not Eligible

Marks: 5

Section D: Challenge Questions (30 Marks)

Q25. Simple Shopping Cart

Create a dictionary representing a shopping cart:

None

```
{  
    "Laptop": 50000,  
    "Mouse": 500,  
    "Keyboard": 1200  
}
```

Calculate:

1. Total bill amount
2. Average price of items

Marks: 10

Q26. Student Grade Evaluator

Accept marks from the user and display:

Marks Range	Grade
-------------	-------

90-100	A
--------	---

75-89	B
-------	---

60-74	C
-------	---

40-59	D
-------	---

Below 40	F
----------	---

Use if-elif-else conditions.

Marks: 10

Q27. Employee Information Manager

Create a dictionary storing:

None

```
{  
    "name": "",  
    "department": "",  
    "salary": 0  
}
```

Perform the following:

1. Accept values from the user.
2. Display employee details.
3. Increase salary by 10%.
4. Display updated salary.

Marks: 10

Bonus Question (Optional – 10 Marks)

Create a small **Student Record System** using a dictionary that stores:

- Student Name
- Roll Number
- Course
- Marks

Display whether the student has passed or failed using conditional statements.